Currently, no purely subtractive system is known (in which addition does not play some role), and all subtractive systems use both addition and subtraction. Most subtractive systems only employ subtraction for 1 (therefore, for example, in such a base-10 subtractive system, the number 38 is written as $30 + 8$, but the number 39 is written as $40 - 1$).

Problem 9.6

Yoruba (Harold Somers, UKLO 2020)

Here are some numbers in Yoruba:

èji	2	*ẹ́ẹ́rìndilogóji*	36
ẹ̀rin	4	*ẹ̀rìndogóji*	44
àrun	5	*àádorin*	70
ẹ̀rinlá	14	*ẹ́ẹ́tàdilogórin*	77
eéjìdilogun	18	*ẹ̀tàdogórin*	83

Problem 9.6a Write in numerals:

a. *àádota*
b. *àrùndogórin*
c. *aárùndilogórin*
d. *ẹ̀tàdogórun*
e. *òkándilogóji*

Problem 9.6b Write in Yoruba: 12, 45, 57, 90, 99.



The marks above the vowels denote tones; *e* and *ẹ* are distinct vowels.

Solution

Firstly, we can notice the pair 4 and 14, in which the only difference is the suffix *-lá*. We can therefore deduce that $10 + X = X$*-lá*, so this system is most likely base-10.

Next, we have the pair 36 and 44, in which the only difference is *-il-*. Moreover, the first part of the word is highly similar to the word for 4 (in the case of 44 it is actually identical, while in the case of 36 it is slightly mutated), and the